

All-Inside Quadrupled Semitendinosus Autograft Shows Stability Equivalent to Patellar Tendon Autograft Anterior Cruciate Ligament Reconstruction: Randomized Controlled Trial in Athletes 24 Years or Younger



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Purpose: To compare clinical outcomes of knee anterior cruciate ligament (ACL) autograft reconstruction using all-inside quadrupled semitendinosus (AIST) versus bone–patellar tendon–bone (BPTB) in a high-risk athletic population 24 years or younger. **Methods:** Skeletally mature candidates younger than 24 years old with an ACL tear were randomized into either the AIST ($n = 32$) or BPTB ($n = 32$) group and were followed for 2 years. Magnetic resonance imaging scans were obtained at 1-year follow-up, and radiographs were obtained at 2-year follow-up. All surgeries were performed by a single surgeon using an anteromedial portal to establish the femoral tunnel. The primary outcome measure was KT-1000 stability testing. Secondary outcome measures included International Knee Documentation Committee (IKDC) Knee Evaluation Form, IKDC Subjective Form, Knee Injury and Osteoarthritis Outcome Score, Marx Activity Scale, visual analog pain scale, and SF-12 (Mental and Physical). Results: At 2-year follow-up, no statistical difference existed with KT-1000-measured side-to-side laxity between AIST (0.3 ± 0.7 mm, 95% confidence interval 0.0–1.0 mm) and BPTB (0.0 ± 0.8 mm, confidence interval CI -0.3 to 1.1 mm) ($P = .197$). In addition, no statistical differences between the groups were found for IKDC Subjective Form, Knee Injury and Osteoarthritis Outcome Score, Marx, SF-12 Mental, SF-12 Physical, or with regards to imaging findings. Patients with BPTB reported significantly greater postoperative pain scores at days 2 ($P = .049$), 3 ($P = .004$), and 7 ($P = .015$) and had significantly greater kneeling pain at 2 years ($P < .019$). A return to sport questionnaire at 2 years revealed no significant difference between the groups for returning to preoperative level of sport activity (83% AIST, 74% BPTB; $P = .415$). Two graft retears (7%; $P = .222$) occurred in the AIST group. Three patients in the BPTB group experienced ACL tears in the contralateral knee (9%; $P = .239$). **Conclusions:** ACL reconstruction with an all-inside quadrupled semitendinosus autograft construct is equivalent to patellar BPTB autograft based on KT-1000 stability testing in athletes 24 years or younger. **Level of Evidence:** randomized controlled trial with 92% 2-year follow-up, Level I.

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The authors report the following potential conflict of interest or source of funding: P.A.S. reports grants from Arthrex, during the conduct of the study; personal fees from Arthrex; and stock ownership in Spinal Simplicity, outside the submitted work. Full ICMJE author disclosure forms are available for this article online, as [supplementary material](#).

Received April 5, 2019; accepted January 18, 2020.

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0749-8063/19446/\$36.00

<https://doi.org/10.1016/j.arthro.2020.01.048>

Anterior cruciate ligament (ACL) injury can have far-reaching sequelae from resultant instability causing functional limitations, meniscal tearing, and premature arthritic development.^{1–7} The optimal ACL reconstruction technique has not been determined, especially in young active patients, with reported graft re-tear rates ranging from 10% to 25%.^{8–13} This is a significant clinical problem due both to the increasing frequency of ACL injuries occurring in this group^{14–17} and to the high rate of secondary injuries following ACL reconstruction in this particular cohort.^{8–11,18–23} A recent systematic review and meta-analysis found that nearly 1 in 4 young athletes who undergo ACL reconstruction and return to high-risk sport will sustain

another ACL injury during their career.²³ Therefore, if 1 ACL graft showed better clinical outcomes in the challenging case of young patients with ACL tears, that would be important for clinicians with respect to surgical decision making.

Hamstring and bone–patellar tendon–bone (BPTB) autografts are the most common grafts used for ACL reconstruction.^{24–27} Previous studies of hamstring autograft versus patellar tendon BPTB autografts have showed comparable outcomes (Table 1).^{1,3,28–47} Certainly, it should be recognized that in these previous studies, there is variability with overall patient age, operative technique and graft fixation, representing confounding variables affecting outcomes. However, overall, BPTB grafts have generally been shown to have a lower risk of graft failure compared with hamstring tendon grafts,^{33,48–51} to be more stable than hamstring tendon grafts,^{44,48,52,53} and are considered by many as the favored graft in young and active patients.^{54,55} However, BPTB grafts have been associated with significant increases in knee pain⁵⁶ compared with hamstring grafts.^{1,3,28,32,39,41,56–61} This adverse development after BPTB reconstruction in a younger population could outweigh the benefits of improved stability, but there has not been a randomized controlled trial of BPTB versus hamstring conducted in patients less than 25 years old to help clinicians make decisions regarding advantages and disadvantages of graft choice in this challenging population.

All-inside ACL reconstruction is based on a tibial socket instead of a full tunnel.⁶² This approach has potential advantages from a patient, biologic, and stability standpoint. Lubowitz et al.⁶³ showed less pain with all-inside allograft ACL reconstruction compared with a full tibial tunnel in a randomized controlled trial. Smith et al.⁶⁴ found in a translational canine model comparing graft healing of all-inside fixation versus interference screw fixation, restoration of normal 4-zone histologic graft integration only with an all-inside socket. Biomechanically, the all-inside surgical technique consists of suspensory fixation on both the femoral and tibial side, allowing for intraoperative graft retensioning.^{65,66} Therefore, this all-inside approach could improve graft integration and stability of a hamstring construct,⁶⁷ as opposed to other described hamstring reconstruction techniques that have been compared with a BPTB graft, along with the added benefit of less anterior knee pain.

The purpose then of this investigation was to compare knee ACL autograft reconstruction using all-inside quadrupled semitendinosus (AIST) versus BPTB in a high-risk athletic population 24 years or younger. Although some research has reported no difference in knee stability between hamstring tendon and BPTB, other research has reported differences that exist between hamstring tendon and BPTB in terms of knee stability. We do believe that the AIST method has

potential advantages that might account for the improved results, which we have included in the discussion section. A design goal of this single-surgeon study was to standardize operative technique and control potential confounding variables as much as possible. Given the importance of restoring joint stability with ACL reconstruction for an optimal outcome, our null hypothesis was that there would be no difference in primary outcome KT-1000 stability at 2-year follow-up between these 2 groups.

Methods

After a prospective power analysis was performed and following approval by an institutional review board (Salus IRB, formerly RCRC), patients who were seeking surgical intervention for an ACL tear were assessed for eligibility into the study.

Eligibility Criteria

Eligibility for the study was assessed before consent and conducted by a research coordinator at the study site. The inclusion criteria were skeletally mature (Tanner 5) patients 24 years or younger with an ACL-deficient knee who agreed to ACL reconstructive surgery using autograft tissue. Patients with associated meniscal and chondral pathology (except those meeting exclusion criteria) were included in the study, with such pathology being treated at the time of ACL reconstruction at the discretion of the study surgeon. All pathology and treatments were recorded. Excluded were patients who had previous ACL reconstructive surgery, patients who were undergoing multiligament surgery (medial collateral ligament, posterior cruciate ligament, lateral collateral ligament, posteromedial corner, or posterolateral corner repair or reconstruction), patients who were pregnant or nursing at the time of study enrollment, and vulnerable populations (including employees or staff at the study site and athletes on teams on which the principal investigator served as head team physician).

Enrollment and Randomization

Subjects who met the eligibility criteria were consented and randomized into either the AIST or BPTB group. Randomization was conducted by using previously generated envelopes. Both the site research coordinator and primary investigator were blinded to the allocation and randomization process up until opening the envelopes. These envelopes were allocated by using a random number generator to assign patient numbers and study groups. Once these envelopes were received by the study site, they were placed in numbered binders and remained unopened until after the consent process, at which point the envelopes were opened by the research coordinator to determine the randomized study group. Patients then underwent a thorough knee examination by the primary

Table 1. Randomized Controlled Studies Comparing Hamstring Tendon and BPTB

Study	Follow-Up, y	Number of Subjects	Population Average Age	Population Age Range	Hamstring Type	Graft Failure	CL Tear	Note
Aune et al. ²⁸	2	72	26	15-50	STG	HT: 5.4% BPTB: 2.9%	NR	Similar outcomes; ↓ flexion strength HT; ↑ kneeling pain BPTB
Barenius et al. ¹	8	164	NR	15-45	4ST	HT: 2.7% BPTB: 2.6%	HT: 4.0% BPTB: 9.0%	Similar outcomes; ↑ kneeling pain BPTB
Bjornsson et al. ³	16	193	BPTB: 28.2 HT: 26.8	14-59	3ST, 4ST, STG	HT: 8.1% BPTB: 6.6%	HT: 7.0% BPTB: 9.8%	Similar outcomes; ↑ knee pain BPTB; ↑ KOA BPTB
Drogset et al. ²⁹	2	115	27	18-50	STG	HT: 2% BPTB: 2%	NR	Similar outcomes; ↓ flexion strength HT; ↑ anterior knee pain BPTB
Ejerhed et al. ³⁰	2	71	BPTB: 26 HT: 29	14-59	3ST, 4ST	HT: 5.4% BPTB: 2.9%	NR	Similar outcomes; ↑ knee pain BPTB
Eriksson et al. ³¹	2	180	25.7	15-45	4ST	HT: 4.1% BPTB: 2.5%	NR	Similar outcomes; ↓ extension BPTB
Feller and Webster ³²	3	65	BPTB: 25.8 HT: 26.3	18-40	STG	HT: 0% BPTB: 3.2%	NR	↑ laxity HT; ↑ kneeling pain BPTB; ↓ extension BPTB; ↓ flexion strength HT
Harilainen et al. ³³	5	99	Female: 33 Male: 29	NR	STG	HT: 8.3% BPTB: 0%	HT: 8.3% BPTB: 2%	Similar outcomes; ↑ tunnel enlargement HT
Holm et al. ³⁴	10	72	26	15-50	STG	HT: 10.3% BPTB: 10.7%	HT: 13.8% BPTB: 10.7%	Similar outcomes; ↑ KOA BPTB
Ibrahim et al. ³⁵	5	85	22.3	17-34	STG	NR	NR	Similar outcomes; ↑ patellofemoral pain BPTB; ↑ loss of motion BPTB
Jansson et al. ³⁶	2	99	NR	NR	STG	HT: 4% BPTB: 0%	NR	Similar outcomes
Kautzner et al. ³⁷	2	150	26	17-47	STG	HT: 5.5% BPTB: 4.1%	NR	Similar outcomes; ↑ anterior knee pain BPTB first 6 months
Konrads et al. ³⁸	10	62	29.8	18-44	3ST, 4ST	HT: 3.2% BPTB: 0%	NR	Similar outcomes
Laxdal et al. ³⁹	2	134	BPTB: 28 HT: 25	12-52	3ST, STG	HT: 5.1% BPTB: 2.5%	HT: 0% BPTB: 2.5%	Similar outcomes; ↑ kneeling pain BPTB
Lidén et al. ⁴⁰	7	71	BPTB: 28 HT: 29	14-59	3ST, 4ST	HT: 5.4% BPTB: 5.9%	NR	Similar outcomes
Maletis et al. ⁴¹	2	99	BPTB: 27.2 HT: 27.7	14-48	STG	HT: 1.9% BPTB: 0%	HT: 1.9% BPTB: 4.3	Similar outcomes; ↓ flexion strength and ↑ difficulty jumping HT; ↑ return to previous Tegner level BPTB; ↓ extension BPTB; ↑ kneeling pain BPTB; ↑ sensory deficit BPTB
Sajovic et al. ⁴²	5	64	BPTB: 27 HT: 24	14-46	STG	HT: 7% BPTB: 8%	HT: 7.1% BPTB: 11.5%	Similar outcomes; ↑ KOA BPTB
Sajovic et al. ⁴³	11	64	BPTB: 27 HT: 24	14-46	STG	HT: 6% BPTB: 12%	HT: 7.1% BPTB: 11.5%	Similar outcomes; ↑ presence of pivot shift (1+) BPTB; ↑ KOA BPTB
Sajovic et al. ⁴⁴	17	64	BPTB: 27 HT: 24	14-46	STG	HT: 6.3% BPTB: 9.4%	HT: 9.4% BPTB: 9.4%	Similar outcomes; ↑ laxity HT; ↑ KOA BPTB
Shaieb et al. ⁴⁵	2	82	BPTB: 32 HT: 30	14-53	STG	HT: 5.4% BPTB: 6.1%	HT: 8.1% BPTB: 9.1%	Similar outcomes; ↑ loss of motion and patellofemoral pain BPTB; ↑ KOA BPTB
Taylor et al. ⁴⁶	2	64	BPTB: 21.7 HT: 22.1	17-44	STG	HT: 12.5% BPTB: 9.4%	HT: 0% BPTB: 6.3%	Similar outcomes; ↑ Tegner level BPTB
Webster et al. ⁴⁷	15	65	BPTB: 26.6 HT: 26.1	18-40	STG	HT: 8.8% BPTB: 3.2%	HT: 5.9% BPTB: 12.9%	Similar outcomes; ↑ sport participation BPTB

3ST, 3-strand semitendinosus; 4ST, quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; CL, contralateral; HT, hamstring tendon; KOA, knee osteoarthritis; NR, not recorded; STG, semitendinosus + gracilis.

2yr Post Op Patient Return to Sport Evaluation

Patient Instructions: Please answer the following questions based on your current level of ability in sports. If a question does not apply, please list N/A in the blank.

- 1) Are you playing the same sport(s) that you participated in before you had the knee injury?

☐ Yes ☐ No

- 2) What level of sports are you participating in?

☐ Recreational
☐ High School Level
☐ College Level
☐ I am not currently participating in sports

- 3) What sports are you playing now?

- 4) At what level now, in comparison to before your knee injury, do you feel you are back playing at?

☐ 100%
☐ 75%
☐ 50%
☐ 25%
☐ I am unable to play

- 5) If you are unable to play, what sports are you unable to play now?

- 6) If you are not playing at 100% of your pre-injury level of play, what is the **primary** reason?

☐ I do not trust my knee like I used to
☐ I still have pain when I play
☐ I seem to have reduced range of motion
☐ I do not feel strong enough
☐ Psychologically, I do not feel ready or have a fear of re-injuring my knee
☐ Other reason? Explain: _____

- 7) If you are not playing at 100% of your pre-injury level of play, what are the additional reasons? Check all that apply.

☐ I do not trust my knee like I used to
☐ I still have pain when I play
☐ I seem to have reduced range of motion
☐ I do not feel strong enough
☐ Psychologically, I do not feel ready or have a fear of re-injuring my knee

Thank You!

Return to Sport

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Fig 1. Return to sport (RTS) questionnaire.

investigator, including preoperative KT-1000 (MED-metric Corporation, San Diego, CA) testing.

Outcome Measures

KT-1000 manual maximum side-to-side measured laxity was the primary outcome measure. Secondary outcome measures included the International Knee Documentation Committee (IKDC) Knee Evaluation Form, Subjective IKDC Knee Evaluation Form, Knee Injury and Osteoarthritis Outcome Score (KOOS), Marx Activity Scale, SF-12, and patient-reported pain visual analog scale (VAS). Outcomes were measured before surgery and at 6 weeks, 6 months, 12 months, and 24-month follow-up visits. VAS was specifically collected more frequently postoperatively at 1, 2, 3, 7, 14, 21, and 28 days. These outcome measures were all collected in office, except for the postoperative VAS, which was sent home with the patient following surgery and brought back to the clinic at their 6-week appointment. Participants also filled out a return-to-sport questionnaire at 2-year follow-up (Fig 1).

Imaging and Additional Procedures

Magnetic resonance imaging (MRI) studies were done at 1 year postoperatively. Radiographs were done at

24-month follow-up. Any additional knee injury, surgery, or complication was documented.

Surgical Technique

All-Inside Quadrupled Semitendinosus

The GraftLink technique (Arthrex, Naples, FL) uses a single semitendinosus tendon quadrupled with suspensory fixation on both the femur and tibia.^{68,69} The semitendinosus tendon was harvested from a single-incision posterior approach.^{70,71} A graft preparation station (Arthrex) was used, placing a TightRope Reverse Tension (RT) suspensory button (Arthrex) on the femoral side of the graft, and an Open TightRope Attachable Button System (ABS) suspensory loop (Arthrex) on the tibial side. The tendon graft was passed through the respective TightRope loops and then quadrupled, and the 2 free ends on the tibial side were sutured together with #0 FiberLoop (Arthrex) in a SpeedWhip type pattern leaving these suture ends intact.^{72,73} The graft was then placed under 75 Newtons of tension, and the 4 tendon limbs on each side of the respective TightRope loops were sutured together with 2 #0 FiberWire sutures passing through each limb linking them in this fashion creating the final graft

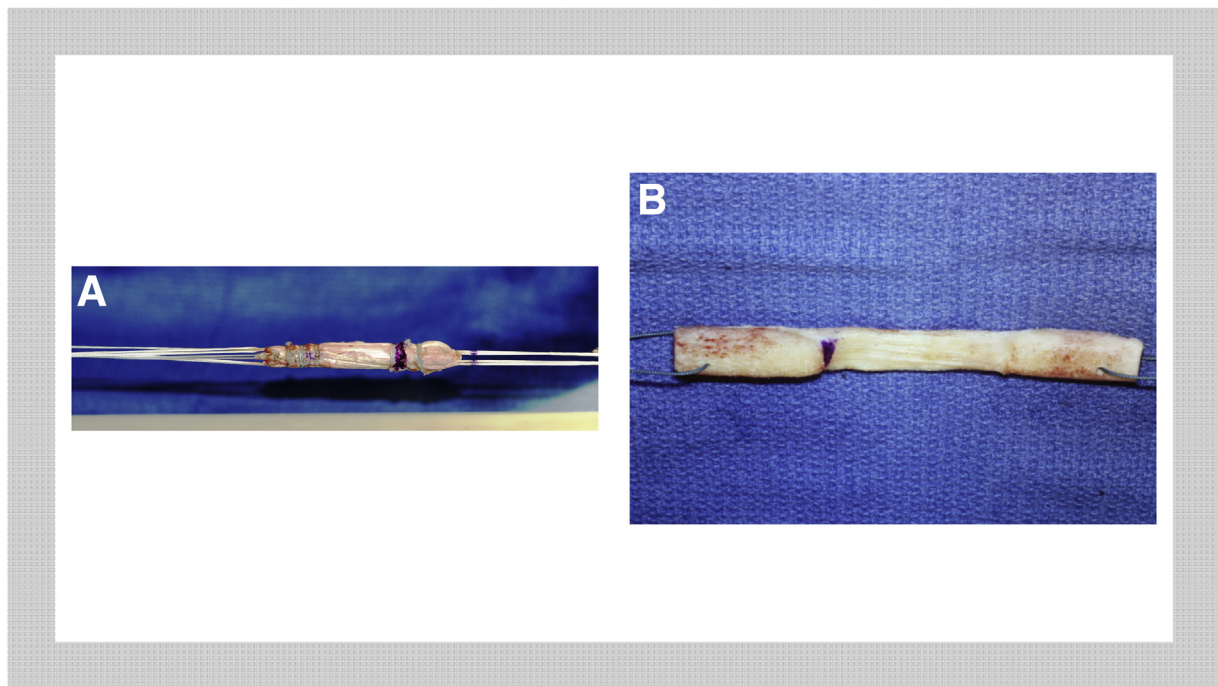


Fig 2. (A) The quadrupled semitendinosus GraftLink construct, secured by #2 FiberWire cerclage sutures on each side of the suspensory fixation loops. (B) Patellar tendon graft with tibial and patellar bone plugs and #5 Ethibond passing sutures in place.

construct (Fig 2). Graft diameters were then measured on a sizing block post construction.

In no cases was a notchplasty done. Referencing off any remaining native ACL fibers and the over the top position with a guide (Arthrex), a pin was placed and drilled out the lateral femur. In this way, the femoral socket was created as close as possible to the anatomic ACL center via anteromedial portal drilling via a low-profile reamer (Arthrex) matching the graft diameter. Minimal graft in socket depth was 20 mm, but the femoral socket was overreamed by 3 to 4 mm to allow for graft retensioning. The pin was used to place a #2 FiberWire suture (Arthrex) shuttle suture here. The intraarticular ACL graft distance was measured with an intraarticular ruler (Arthrex), and this distance was added to the length of the graft in the femoral socket to determine the depth of graft in the tibial socket based on the total graft length. The tibial socket was created also at the anatomic tibial site indexing off the anterior horn of the lateral meniscus using a FlipCutter aiming guide (Arthrex). A straight FlipCutter pin matching the graft diameter was then drilled into the joint and “unflipped” to become a reamer to retrocut the tibial socket, which was also reamed 5 mm deeper than needed to allow for optimal graft tensioning with the all-inside technique. The FlipCutter pin was drilled back into the joint and “unflipped” and removed and a #2 TigerStick suture (Arthrex) was passed through this FlipCutter pin hole up into the joint and retrieved as a tibial shuttle suture. The #2 FiberWire shuttle suture from the femoral socket was used to pull the graft

through the anteromedial portal across the joint up into the femur, flipping the suspensory TightRope RT button on the lateral femoral cortex. The graft was then hoisted up into the socket to the appropriate depth with the TightRope shortening strands. The tibial portion of the graft with the attached Open TightRope ABS loop and the 2 ends of the O FiberWire suture at the tibial end of the graft were then passed via the TigerStick shuttle suture into the tibial socket. An 11-mm TightRope ABS button was applied to the Open TightRope ABS loop passing the O FiberWire strands from the end of the graft through the button slots as well. Holding the foot so that the knee was fully extended (or hyperextended if present), the tibial shortening strands were secured to the button for graft fixation. The knee was then vigorously cycled at least 25 times, and always with the knee in full extension (or hyperextension if present) the shortening strands first on the femur were retensioned, and then the shortening strands on the tibia were then retensioned. The tibial shortening strands were then tied to the button here, as were the O FiberWire strands from the graft itself for additional fixation.

Bone—Patellar Tendon—Bone

Patellar tendon graft harvest began with a midline incision over the patellar tendon to expose the paratenon, which was incised to expose the underlying patellar tendon. After tendon width was measured, the middle third of the tendon was marked, and parallel knife cuts were made from the inferior pole down over

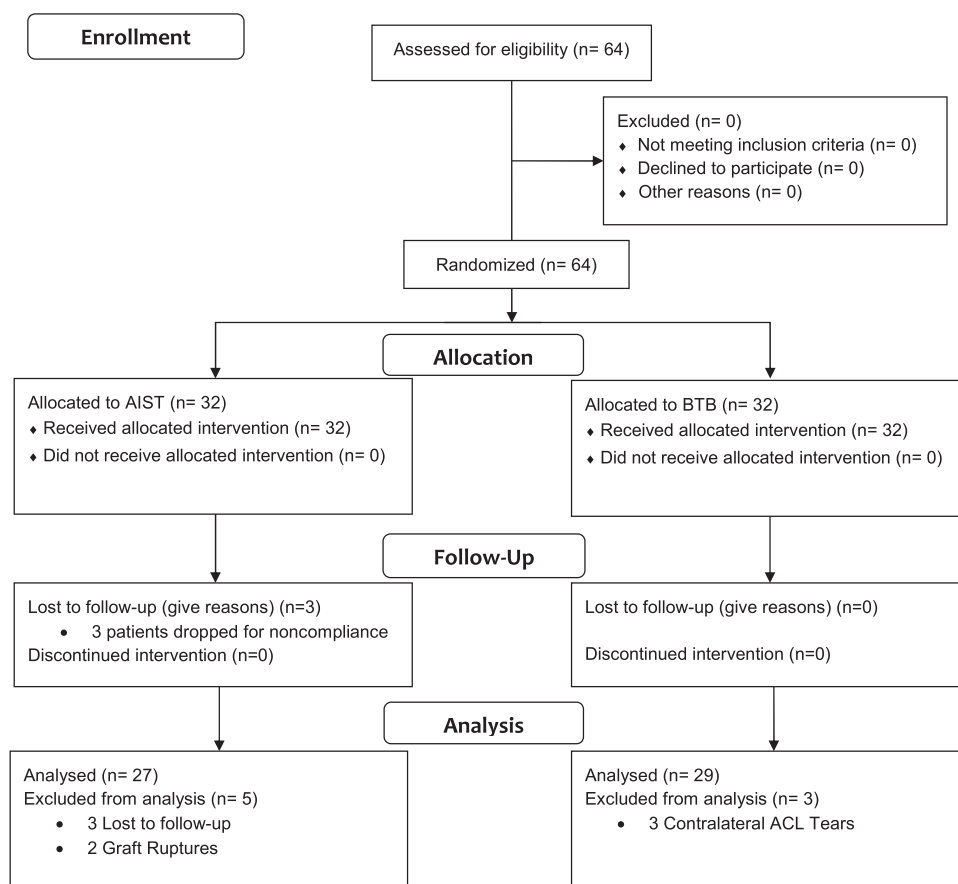


Fig 3. Flowchart of 64 participants in this randomized controlled trial. (ACL, anterior cruciate ligament; AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone.)

the proximal tibia. An oscillating saw was used to harvest the bone plug on the tibial side, which was either 10 or 11 mm in width depending on graft width, followed by harvesting a more triangular shallower bone plug from the patella the same width. Bone grafting of both the tibial and patellar defects was done followed by wound closure in layers. The tibial bone plug was routinely passed into the femur, and both plugs were fashioned to fit either the 10 mm or 11 mm size as appropriate. Drill holes in the plugs were made for placement of #5 Ethibond sutures (Johnson & Johnson, New Brunswick, NJ) for graft passage. Graft thickness was measured with a caliper.

As per the AIST cohort, a notchplasty was not done in any case. The femoral socket was created the same way placing a guide pin at the anatomic ACL center through the anteromedial portal, and then reaming with a low-profile reamer matching the bone plug diameter. A #5 Ethibond femoral socket shuttle suture was placed via the guide pin. The tibial tunnel was created retrograde from within the joint positioning either a FlipCutter or RetroCutter (Arthrex) matching bone plug size at the anatomic ACL tibial attachment. The graft was passed retrieving the #5 Ethibond shuttle suture from the femoral socket through the tibial tunnel with the tibial

plug oriented with the cancellous side anteriorly as it was pulled into the femoral socket. Through the anteromedial portal, a nitinol wire was passed on top of the bone plug followed by a tap matching screw diameter, and then fixation with a Biocomposite interference screw (Arthrex) most commonly 7 mm in diameter and 23 mm long. On the tibial side, the patellar bone plug was rotated externally 90° and a nitinol wire passed above the plug into the joint. After tapping, fixation was done with a Biocomposite interference screw (Arthrex) usually 10 mm in diameter and 23 mm long to match the patellar bone plug length. Tibial graft fixation was always done holding the foot so that the knee was in full extension (or hyperextension if present).

Rehabilitation Protocol

The rehabilitation protocol for both the AIST and BPTB cohorts were the same. Beginning the day of surgery, early motion was instituted with use of a continuous passive motion machine (Flex-Mate K500; KLC Services, Hillard, OH). Supervised physical therapy was started the day after surgery focusing on quadriceps sets, straight leg raises, patellar mobilization and active flexion. Toe touch weightbearing was instituted until

Table 2. Demographic Data, Intraoperative Findings, and Baseline Patient Data

	AIST (N = 32)	BPTB (N = 32)
Patient demographics		
Patient age, y	17.6 ± 2.5	17.8 ± 2.3
Sex (% female)	19 (59%)	17 (53%)
Side of injury (% right)	19 (59%)	16 (50%)
Time to surgery, d	47.1 ± 34.9	46.2 ± 58.2
Baseline patient data		
IKDC objective grade		
A (%)	0	0
B (%)	0 (0%)	1 (3%)
C (%)	14 (44%)	14 (44%)
D (%)	18 (56%)	17 (53%)
KT-1000 Δ Maximum	5.8 ± 1.5	6.3 ± 1.8
IKDC subjective	45.3 ± 17.2	41.0 ± 14.7
KOOS		
Function	53.1 ± 36.9	39.7 ± 36.3
Quality of Life	19.4 ± 11.9	15.6 ± 12.4
Symptoms	58.1 ± 16.2	54.2 ± 15.2
VAS	0.2 ± 0.2	0.2 ± 0.2
MARX	14.6 ± 2.9	14.9 ± 1.8
SF-12		
Physical	39.0 ± 8.0	36.0 ± 5.3
Mental	53.3 ± 9.9	53.4 ± 9.4
Intraoperative findings		
Meniscus tear(s)		
Medial (%)	13 (41%)	6 (19%)
Lateral (%)	7 (22%)	14 (44%)
Any (%)	15 (47%)	16 (50%)
Both (%)	5 (16%)	4 (13%)

AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; VAS, visual analog scale.

patients demonstrated good quadriceps function and leg control. Closed chain strengthening was emphasized. Time to return to play varied pending passage of physical therapy functional testing and physician clearance but was never sooner than 6 months after surgery.

Statistical Methods

Pre-study sample size analysis revealed that a sample size of 24 patients per group was required to detect a clinically relevant difference in KT-1000 knee anteroposterior laxity side-to-side difference (compared with normal, contralateral knee) of 2.5 mm between the 2 groups at 2-year follow-up, assuming a standard deviation of 3.0.⁷¹ Using a *t*-test, alpha value of 0.05, and beta of 0.8, power analysis suggested 24 subjects per group were necessary.⁷⁴ Total enrollment was set at 64 patients, assuring a minimum of 24 patients in each group if 8 patients in each group (< 20%) were lost to follow-up (transfer bias). VAS, Marx Activity, KOOS, IKDC Subjective, and SF-12 secondary outcomes were measured using a 2-way repeated measures analysis of variance (ANOVA) to examine both the effect of time point and group, with further 2-way

repeated-measures ANOVA performed to compare both incidence of kneeling pain and whether the presence of meniscal pathology had any effect on clinical and outcome measures. Two proportion tests were used to compare categorical data between the 2 groups and to compare the percentage of patients who did or did not reach defined minimal clinically important difference (MCID) thresholds, and a *t* test was used to compare radiographic data.

Results

A total of 64 patients (36 female, 28 male) were included in this study, with 92% completing their 2-year follow-up: 32 from the BPTB group and 27 from the AIST group (Fig 3). There were 15 male and 17 female patients in the BPTB group and 13 male and 19 female patients in the AIST group (Table 2). The average age of the entire study population was 17.7 ± 2.4 years (range 13–24 years). Basketball (31), soccer (11) and football (9)—which are 3 of the greatest-risk sports for ACL injury in high school athletes⁷⁵—were the most common sports at the time of injury (Fig 4). Of the 64 patients, 10 were college athletes, 42 were high school athletes, and 12 were recreational athletes. The overall average time from injury to surgery was 47 days (range 11–340 days). Three patients (4.7%) were lost to follow-up and were removed from the study and statistical analyses prior to study completion. There were 2 graft failures (7%) in patients in the AIST group compared with no graft failures in the BPTB group (*P* = .222). Three patients (9%) in the BPTB group experienced contralateral ACL tears compared with no contralateral ACL tears in the AIST group (*P* = .239). These patients who sustained subsequent ACL injuries were included in statistical analyses until the time point of their reinjury, after which they were removed.

Graft Specifications

AIST constructs had an average graft diameter of 8.8 ± 0.62 mm (range 8–10 mm). The average total graft length for this cohort was 68.6 ± 4.0 mm. Graft length in the femur was 22.0 ± 2.1 mm, graft length in the tibia was 28.6 ± 1.7 mm, and intraarticular distance was 23.4 ± 1.4 mm.

BPTB autografts had an average graft width of 11.1 ± 0.80 mm (range 10–13 mm), and a graft thickness of 4.2 ± 0.6 mm. Femoral socket depth averaged 22.8 ± 0.8 mm, matching tibial bone plug length, which averaged 22.0 ± 0.9 mm. Patellar bone plug length was 25.5 ± 2.6 mm.

Associated Surgical Procedures

Overall, there were 31 cases (48%) of meniscal tears in the study population, including 19 (30%) and 21 (33%) tears of the medial and lateral meniscus,

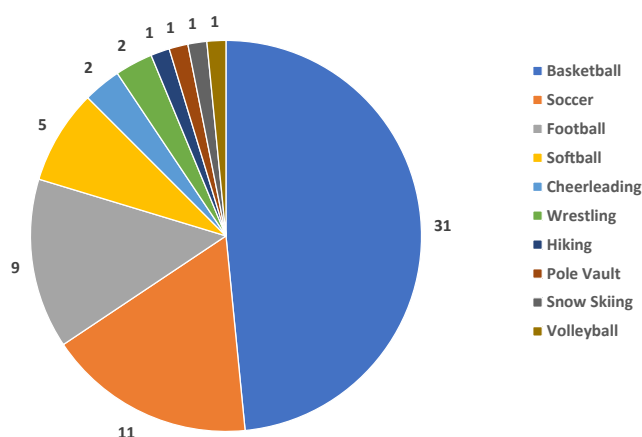


Fig 4. Etiology of anterior cruciate ligament injury of randomized participants.

respectively (Table 3). Tears of both the medial and lateral meniscus occurred in 9 cases (14%).

In the AIST group, there were 13 medial meniscus repairs (41%) and 6 lateral meniscus repairs (19%), so 60% of this group had an associated meniscus repair. Notably, only 1 patient had a partial meniscectomy, which was lateral (3%), and no patient had a partial medial meniscectomy. For the BPTB cohort, there were 7 lateral meniscus repairs (22%) and 6 medial meniscus repairs (19%) for a total of 41% having a meniscus repair done. In the BPTB group, 9 had partial lateral meniscectomy done (28%), but no one had a partial medial meniscectomy.

Relative to other procedures, in the AIST group, there was 1 plica synovectomy (3%), 3 patellar chondroplasties (9%), and 1 medial femoral chondroplasty (3%). In the BPTB group, 1 patient had chondroplasty of the patella (3%) and 1 had a microfracture of the medial femoral condyle (3%).

Two subjects (3%) underwent additional surgeries, with one occurring in each group. One subject in the BPTB group underwent tibial interference screw removal 19 months following reconstruction. One subject in the AIST group reinjured their repaired medial meniscus and had to have a partial medial meniscectomy performed 10.5 months following ACL reconstruction. The 3 BPTB patients with contralateral tears all underwent ACL reconstruction on the contralateral knee, but were still seen at 2-year follow up.

Due to the prevalence of meniscal injury, a 2-way repeated measures ANOVA was performed to determine if the treatment of meniscal pathology affected outcomes. A difference existed in VAS score at day 1 ($P = .003$); however, no other differences existed at the other VAS time points ($P > .092$). In addition, there was no statistical difference with KT-1000 testing ($P = .670$), Marx Activity ($P = .184$), KOOS Symptoms ($P = .858$), KOOS Function ($P = .561$), KOOS Quality of Life ($P = .769$), or IKDC Subjective Score ($P = .871$)

with those patients who had a meniscus repair and/or partial meniscectomy versus those who did not. However, statistical evaluation of the subgroup of partial lateral meniscectomy patients was insufficiently powered.

Complications

There were no postoperative infections in either group. No patient required manipulation or arthroscopic scar tissue debridement for loss of motion.

Outcome Data

KT-1000 arthrometer manual maximum testing found significant differences between AIST versus BPTB at both the 6-week ([BPTB: -0.3 ± 1.1 mm, 95% confidence interval {CI} -0.7 mm to 0.1 mm]; [AIST: 0.4 ± 0.9 mm, 95% CI 0.1 - 0.7 mm]; $P = .009$) and 1-year ([BPTB: -0.3 ± 1.1 mm, 95% CI -0.7 mm to 0.1 mm]; [AIST: 0.9 ± 1.0 mm, 95% CI 0.5 - 1.3 mm]; $P = .001$) time points (Table 4). However, at 2 years, KT-1000 testing showed no difference between AIST (0.3 ± 0.7 mm, 95% CI 0.0 - 0.6 mm) and BPTB (0.0 ± 0.8 mm, 95% CI -0.3 mm to 0.3 mm) ($P = .197$). Keeping consistent with the power analysis, the MCID for KT-1000 testing was analyzed as individuals who presented with a side to side difference of greater than 2.5 mm. No significant differences were found between the groups (Table 5).

No significant statistical differences were seen between either group for the following secondary outcome measures: IKDC Knee Evaluation Form, Marx Activity, KOOS, IKDC Subjective, or SF-12. Secondary outcome measures were further analyzed using a MCID value of 16.7 for IKDC Subjective Score,⁷⁶⁻⁷⁸ a smallest detectable change (SDC) value of 17.4 for KOOS Symptoms⁷⁶, an SDC value of 18.8 for KOOS quality of life,⁷⁶ and a minimal detectable change (MDC) value of 9.9 for Marx Activity.^{79,80} However, the difference of the corresponding secondary outcome measure scores were well below the established thresholds and not statistically different or clinically relevant.

The BPTB group scored significantly greater [6.1 ± 2.9 , 95% CI 5.1 - 7.1], [5.3 ± 2.5 , 95% CI 4.4 - 6.2], [3.9

Table 3. Associated Operative Procedures

Procedure	AIST	BPTB
Medial meniscus repair	13	6
Medial meniscus meniscectomy	0	0
Lateral meniscus repair	6	7
Partial lateral meniscus meniscectomy	1	9
Chondroplasty: patella	3	1
Chondroplasty: MFC	1	0
Plica synovectomy	1	0
Microfracture MFC	0	1

AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; MFC, medial femoral condyle.

Table 4. KT-1000 Measured Laxity and Subjective Outcomes

	6 Weeks	95% CI	P Value	6 Months	95% CI	P Value	1 Year	95% CI	P Value	2 Year	95% CI	P Value
KT-1000 Δ												
Maximum												
AIST	0.4 $\pm$ 0.9	0.3	.009	0.3 $\pm$ 1.2	0.4	.348	0.9 $\pm$ 1.0	0.4	.001	0.3 $\pm$ 0.7	0.3	.197
BPTB	-0.3 $\pm$ 1.1	0.4		0.0 $\pm$ 1.1	0.4		-0.3 $\pm$ 1.1	0.4		0.0 $\pm$ 0.8	0.3	
IKDC Subjective												
AIST	58.5 $\pm$ 12.9	4.8	.22	86.9 $\pm$ 12.0	4.4	.058	91.0 $\pm$ 10.2	3.8	.26	94.8 $\pm$ 6.4	2.4	.248
BPTB	54.3 $\pm$ 12.1	4.2		79.6 $\pm$ 12.2	4.2		86.5 $\pm$ 12.1	4.3		89.4 $\pm$ 9.2	3.3	
KOOS												
Function												
AIST	37.2 $\pm$ 27.0	9.7	.598	79.2 $\pm$ 16.7	6.0	.454	91.2 $\pm$ 14.5	5.3	.219	92.5 $\pm$ 14.0	5.2	.282
BPTB	38.9 $\pm$ 30.6	10.6		72.8 $\pm$ 20.6	7.1		82.5 $\pm$ 15.2	5.3		84.5 $\pm$ 18.7	6.5	
Quality of life												
AIST	39.2 $\pm$ 25.1	9.0	.658	66.3 $\pm$ 27.2	9.7	.183	80.2 $\pm$ 20.6	7.5	.079	83.3 $\pm$ 20.0	7.4	.9
BPTB	36.1 $\pm$ 22.5	7.8		58.4 $\pm$ 21.8	7.6		70.5 $\pm$ 25.0	8.6		82.4 $\pm$ 18.8	6.5	
Symptoms												
AIST	71.8 $\pm$ 15.1	5.4	.411	85.5 $\pm$ 12.5	4.5	.302	87.7 $\pm$ 9.5	3.4	.285	89.7 $\pm$ 7.5	2.8	.348
BPTB	68.9 $\pm$ 16.2	5.6		81.8 $\pm$ 15.1	5.2		84.3 $\pm$ 13.1	4.5		86.6 $\pm$ 13.4	4.6	
SF-12												
Physical												
AIST	39.8 $\pm$ 6.9	2.7	.889	52.6 $\pm$ 7.1	2.8	.457	55.7 $\pm$ 4.5	1.8	.374	55.5 $\pm$ 3.8	1.5	.876
BPTB	39.9 $\pm$ 8.8	3.5		51.3 $\pm$ 7.4	3.0		55.0 $\pm$ 4.7	1.9		56.4 $\pm$ 2.1	0.9	
Mental												
AIST	56.5 $\pm$ 8.9	3.5	.417	58.5 $\pm$ 3.6	1.4	.99	58.2 $\pm$ 3.3	1.3	.99	58.0 $\pm$ 3.7	1.5	.88
BPTB	57.8 $\pm$ 7.0	2.8		57.5 $\pm$ 6.3	2.6		57.6 $\pm$ 4.4	1.8		58.7 $\pm$ 2.2	0.9	
MARX												
AIST	N/A		N/A	N/A		N/A	14.6 $\pm$ 2.9	1.0	.68	13.3 $\pm$ 4.4	1.7	.790
BPTB	N/A			N/A			14.9 $\pm$ 1.8	0.6		13.1 $\pm$ 4.5	1.7	

NOTE. Bolded values represent statistical significance ($P < .05$).

AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; N/A, not applicable.

± 2.4 , 95% CI 3.0–4.8]) than the AIST group ([5.1 $\pm$ 2.5, 95% CI 4.2–6.0], [3.8 $\pm$ 2.3, 95% CI 3.0–4.6], [2.7 $\pm$ 1.8, 95% CI 2.1–3.3]) on postoperative VAS Pain scales at days 2 ($P = .049$), 3 ($P = .004$), and 7 ($P = .015$), respectively (Table 6). However, VAS Pain testing at all other time points showed no significant differences between groups.

Specific questions from the IKDC Subjective (“How does your knee affect your ability to kneel on the front of your knee?”) and KOOS Function (“What degree of difficulty have you experienced during the last week with kneeling in regards to function, sports and recreational activities?”) were analyzed, revealing that the BPTB cohort had significantly greater pain or difficulty at all postoperative time points, with significant differences existing on both IKDC (6 weeks [$P = .003$], 6 months [$P = .001$], 1 year [$P = .001$], 2 years [$P = .005$]), and KOOS (6 months [$P = .044$], 1 year [$P = .01$], 2 year [$P = .019$]) kneeling questions (Fig 5).

There were no significant differences in return to sport rates between the 2 groups (83% AIST; 74% BPTB; $P = .415$). Of those who returned to sport, 75% of the AIST and 59% of the BPTB patients returned to 100% activity ($P = .372$). Four ($n = 4$) patients reported they were unable to return to sports (2 AIST, 2 BPTB). Overall, the most frequent reason given for not returning and playing at 100% of their preinjury level was a lack of trust in the repaired knee for both groups.

One-Year MRI Data

All MRI scans were evaluated by 2 independent radiologists, specifically for graft evaluation, given the nature of the 4 stranded hamstring construct compared with the ribbon BPTB graft. Both radiologists were blinded to the treatment group. All grafts were found to be intact at 1-year follow-up. Graft signal was interpreted as homogeneous or heterogenous, with homogeneous indicating uniform graft healing and

Table 5. KT-1000 MCID (Side to Side Difference >2.5 mm) Results

	Preoperative	6 Weeks	6 Months	1 Year	2 Years
AIST	96.9% (31/32)	9.4% (3/32)	0.0% (0/28)	3.7% (1/27)	3.7% (1/27)
BPTB	96.8 (30/31)	9.4 (3/32)	0.0% (0/32)	0.0% (0/32)	0.0% (0/29)
P value	1.000	1.000	.492	.483	.475

AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; MCID, minimal clinically important difference.

Table 6. VAS Pain Scale Results

Time	AIST	95% CI	BPTB	95% CI	P Value
Day 1	5.5 ± 2.7	1.0	5.7 ± 2.7	0.9	.781
Day 2	5.1 ± 2.5	0.9	6.1 ± 2.9	1.0	.049
Day 3	3.8 ± 2.3	0.8	5.3 ± 2.5	0.9	.004
Day 7	2.7 ± 1.8	0.6	3.9 ± 2.4	0.9	.015
Day 14	1.7 ± 1.6	0.6	2.6 ± 2.2	0.8	.077
Day 21	1.3 ± 1.2	0.5	1.5 ± 2.1	0.7	.786
Day 28	0.8 ± 1.1	0.4	1.2 ± 2.0	0.7	.505
6 weeks	0.4 ± 0.8	0.3	0.5 ± 1.5	0.5	.814
6 months	0.3 ± 0.6	0.2	0.2 ± 0.4	0.1	.765
1 year	0.3 ± 1.3	0.5	0.4 ± 0.9	0.3	.792
2 year	0.1 ± 0.3	0.1	0.3 ± 0.6	0.2	.935

NOTE. Bolded values represent statistical significance ($P < .05$).
 AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; CI, confidence interval; VAS, visual analog scale.

maturation and heterogeneity being indicative of possible granulation tissue or sprain. AIST (28 scans) had 21 homogeneous and 7 heterogenous gradings. BPTB (32 scans) had 27 homogeneous and 5 heterogenous gradings ($P = .367$). One AIST patient did not have an MRI due to a medical contraindication (see Appendix Table 1, available at www.arthroscopyjournal.org).

Two-Year Radiographic Data

Radiographs done at 2 years were analyzed for joint space narrowing or evidence of tunnel widening or cystic change. One patient (BPTB) was felt to have early lateral joint space narrowing. No significant tunnel widening was seen in either group (see Appendix Table 2, available at www.arthroscopyjournal.org).

Discussion

In athletic patients aged 24 years or younger, an all-inside quadrupled semitendinosus autograft using suspensory fixation was found to be comparable with a BPTB autograft with respect to the primary study outcome, measured knee laxity. There were no significant differences found with subjective outcome measures, and post-hoc statistical analyses found no clinically significant differences between the 2 groups for MCID, SDC, or MDC established values. Given the high incidence of ACL tears in young athletic patients,^{15,81,82} as well as the known greater risks of secondary injuries following ACL reconstruction in this group,^{8-11,18-23} this study could help guide ACL treatment and serve as a point of reference for additional needed randomized controlled trials in this important young population.

Our primary outcome was KT-1000 stability at 2 years. Analysis of the KT-1000 data revealed the BPTB group had significantly less measured laxity at 6 weeks and 1 year compared with AIST, but not at 2 years. A possible explanation for the BPTB group being tighter at

1 year that these patients were tighter in extension with motion loss, thereby dampening KT-1000 laxity, but on analysis, there was no significant difference between the groups with range of motion or 1 year MRI analysis, which did not show increased signs of anterior arthrofibrosis about the grafts in the BPTB group. Also, one additional explanation could be related to MRI graft signal being more heterogenous in the hamstring group, representing perhaps a slower revascularization and remodeling process with the 4-stranded graft.^{83,84} However, although there was a trend toward more homogeneous signal grafts with the BPTB group at 1 year, this was not statistically significant. Certainly, the 2-year KT-1000 data for both groups (0.3 ± 0.7 mm AIST, 0.0 ± 0.8 mm BPTB) was well within the accepted 3 mm side to side difference for normal range.⁸⁵ Furthermore, the mean KT-1000 laxity measurements for both groups were close to 0, which could be indicative of improved stability with the studied all inside hamstring construct compared to other hamstring graft constructs that have been reported in

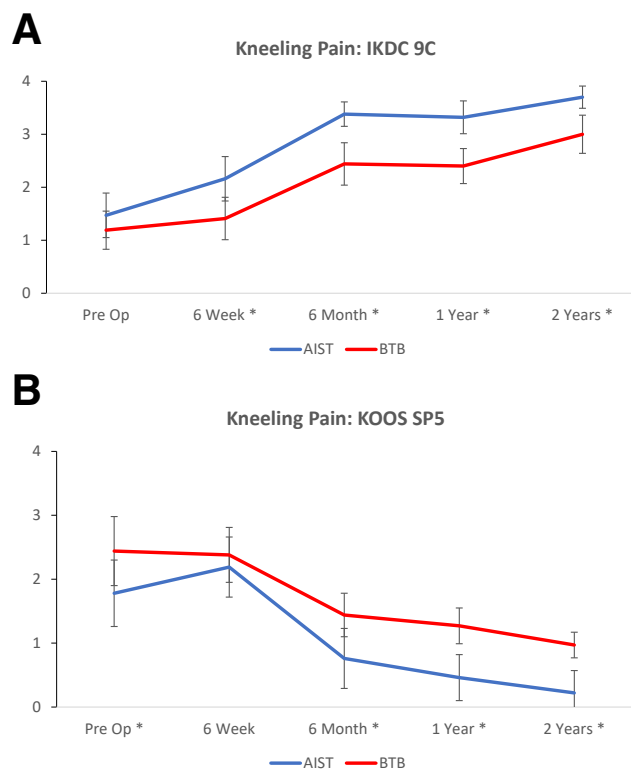


Fig 5. Outcome results displaying 95% confidence intervals for (A) IKDC and (B) KOOS questions pertaining to pain while kneeling. A greater value on the IKDC is indicative of better outcomes, whereas a lower value on the KOOS indicates a better outcome. *Significant difference between groups ($P < .05$). (AIST, all-inside quadrupled semitendinosus; BPTB, bone–patellar tendon–bone; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score.)

the literature as having greater reported laxity measurements).^{30,41,45,86,87}

BPTB is still considered by many to be the favored graft type in patients with generalized joint laxity,^{48,88} young and active patients,^{54,55} and patients with concomitant medial collateral ligament injury.^{89,90} The current study found patients with BPTB had significantly more postoperative pain for the first week after reconstruction, which correlates with Okoroha et al.,⁹¹ who found a significant increase in acute postoperative pain when comparing BPTB with hamstring tendon. Notably, an increase in kneeling pain has been reported in multiple studies and systematic reviews comparing BPTB with hamstring ACL reconstruction.^{1,3,28,32,39,41,56-61} However, these studies have reported on an older group of patients with kneeling pain following BPTB ACL reconstruction. In this study, based on specific outcome questions from IKDC and KOOS, anterior knee pain was significantly increased in the BPTB group. It is impactful that this study now confirms a greater risk for kneeling pain following BPTB ACL reconstruction even in young patients, which represents a very important counseling point in younger patients regarding ACL reconstruction graft choice.

The 2 AIST retears (7%) in this study compare favorably with what has been reported in the literature for hamstring autograft retear rates in young patients.^{11,50,55,92,93} The 2 AIST retears both occurred late in the second year of the study at 20 and 22 months. Their graft diameter was 8 mm and 9 mm, respectively, neither being less than 8 mm in diameter, which has been reported to be a risk factor for hamstring ACL graft retears.⁹⁴⁻⁹⁸ Notably, both of these athletes reinjured while playing basketball, one male (16 years) and the other female (18 years).

A possible explanation for the relatively low retear rate in this study for AIST group compared to other described hamstring ACL reconstructions could related to the all-inside technique itself with adjustable loop suspensory fixation on both the femoral and tibial side allowing for graft retensioning and therefore, a more stable intraoperative construct. Specifically, Noonan et al.⁶⁵ recently reported 0.53 mm time zero slack and 62% tension loss comparing tibial interference screw fixation with and without adjustable loop femoral fixation in an experimental model. This retensioning feature would potentially help explain the low KT-1000 side to side difference for the AIST group at just 0.3 ± 0.7 mm, which was comparable with the BPTB group at 0.0 ± 0.8 mm and again, represented improved objective stability compared to other studies of hamstring grafts reporting actual laxity measurements.^{30,41,45,86,87} Another explanation for the low AIST group retear rates in this study is that none of the grafts had a graft

diameter less than 8 mm, which, as stated previously, has been reported as a risk factor for graft retear.⁹⁴⁻⁹⁸

Another inherent advantage of the AIST technique compared with other hamstring ACL reconstructions is that only the semitendinosus is harvested to create the construct, as many reported studies on hamstring ACL reconstruction have used both the semitendinosus and the gracilis to create the construct (Table 6). Gracilis and semitendinosus harvest is associated with decreases in hamstring flexion strength.^{29,99-102} Furthermore, saving the gracilis potentially lessens the impact of the loss of the dynamic function the hamstrings have in protecting the ACL.¹⁰³⁻¹⁰⁹

It should be recognized that the BPTB cohort in this study overall did well. Graft stability was excellent based on the KT-1000 testing, and there were no graft retears in the BPTB group, which is consistent with recent registry data^{49,50,110,111} and other research^{48,51,112,113} that found a greater retear rate for hamstring autografts compared with BPTB autografts in ACL reconstruction. Notably, the majority of such published studies had a greater overall average age and range than this study. The only 2 randomized controlled studies using only quadrupled semitendinosus revealed a hamstring versus BPTB retear rate of 4.1% versus 2.5%³¹ and 2.7% vs 2.6%¹; but again, these studies included older patients.

While graft retear is an extremely unfavorable event, tears of the contralateral ACL are devastating as well, occurring as or more frequently than graft retears.^{8,13,114,115} Male and female patients younger than 20 years are also 3 times more likely than individuals older than 20 to experience a contralateral ACL tear following ACL reconstruction.¹¹⁶ In this study, there were no contralateral tears in the AIST group and 3 in the BPTB group. Similar to this study, other studies comparing hamstring and BPTB autografts have documented an increased contralateral tear rate with BPTB autografts compared with hamstring tendon autografts.^{8,46,47,117,118} The 3 BPTB contralateral tears occurred in 2 males and 1 female at an average of 10.5 months (9, 10, and 12.5 months respectively) from surgery and occurred while playing football, basketball and soccer. A contralateral ACL tear in a patient under the age of 25 who ends up undergoing bilateral ACL reconstructions is certainly not benign, especially when considering the known risk of knee osteoarthritis development over time following ACL injury even with reconstruction.^{1,3,7,33,34,42-44,57} Furthermore, a contralateral leg injury can be more damaging to an athlete returning to sport than a graft rupture in a reconstructed ACL.¹¹⁹ In a recent case series, Webster et al.¹²⁰ found that return to sport rates in individuals who have undergone bilateral ACL reconstruction averaged only 40%, which is half the return rate of

individuals returning to sport after their first ACL reconstruction.

Our return to sport rate of 83% (AIST) and 74% (BPTB) and return to previous activity level rate of 75% (AIST) and 59% (BPTB) compare favorably with recent return-to-sport research.^{13,121-124} The most frequent reason patients in this study reported not returning to their previous level of sport related to a lack of confidence in their knee, highlighting the significance of psychological readiness in returning to sport and competition.¹²⁵⁻¹³¹

Our follow-up radiographic data showed all grafts were intact on MRI at 1 year. Graft signal interpreted by 2 independent radiologists showed no statistical difference in dark versus heterogenous signal with the two groups. Radiographs at 2 years showed no significant difference in joint space narrowing or tunnel widening, which is similar to other 2-year evidence.^{31,36,46,82}

Limitations

One limitation of the study is that due to the large sample sizes required to confidently compare small changes in the included secondary outcome measures, our included measures could potentially be a result of a type II or beta error. However, the large sample sizes required would be unreasonable for a single center randomized controlled trial. Furthermore, the differences between the AIST and BPTB groups were not found to be statistically significant. In fact, the subjective outcome values were closely correlated, and the differences between the groups were well below the established MCID/SDC/MDC for clinical significance. Similarly, some of the recorded standard deviations for the secondary patient reported outcome measures are high. However, these values are comparable with other randomized control trials that report standard deviation^{28,34}; furthermore, these scores were analyzed for clinical significance with established MCID thresholds when available, and no significant differences were found between the groups.

A second potential limitation of this study was that it was a single center study comparing 2 different fixation methods. However, one senior sports medicine trained surgeon did all the ACL reconstructions in a consistent fashion with the only difference being the graft used and its necessary fixation to help avoid confounding variables, such as different graft placement, graft tensioning or graft fixation. Furthermore, using KT-1000 as a primary outcome compared to a patient reported outcome measure could be viewed as a potential limitation since the power analysis revealed a side to side difference measurement of 2.5 mm as being significant for the purposes of this study, and this study found a side to side difference of 1.0 mm. However, those studies analyzed hamstring grafts that included both semitendinosus and gracilis tendons and used

different fixation methods, potentially explaining this discrepancy. Furthermore, the differences between the 2 groups in both KT-1000 measurements and patient-reported outcome measures for both cohorts fell below established MCID thresholds, which would not affect the conclusions of this study.

Another limitation of the study is that strength data was not collected. It would have been noteworthy to compare quadriceps and hamstring strength between the 2 groups isokinetically to determine the impact of patellar tendon harvest on quadriceps strength, versus semitendinosus harvest and its effect on hamstring strength. However, isokinetic testing was not readily available. The data on objective quadriceps strength would particularly have been helpful analyzing the patients with contralateral ACL tear in the BPTB cohort to assess whether significant quadriceps weakness on the operated side could have played a role for an athlete than favoring this knee returning to play, and potentially overloading the other knee. Similarly, results on functional testing of the 2 cohorts used from the physician counseling standpoint regarding clearance to return to sport was not reportable since patients were seen by multiple physical therapists, so there was a lack of a uniform protocol on functional testing. Furthermore, it was not feasible to collect the data from multiple therapy clinics.

Our focus with the 1-year MRI studies was to assess graft signal integrity by comparing AIST to BPTB, given their different geometries. At that time frame, specifically in this young patient population with good clinical ACL graft stability, we did not suspect frank degenerative joint changes. Two-year radiographs did not show obvious degenerative changes in either group, but not analyzing for articular changes on MRI could be considered a limitation. In addition, the radiologists were blinded to each treatment group, but given the presence of interference screws on MRI for the BPTB group not seen for AIST group, it would have been impossible for them to be completely blinded, which also is a potential limitation.

The 3 patients that were lost to follow-up were all in the AIST group, thus creating a potential limitation. Adding them to the 2 patients with AIST who reinjured their graft meant the AIST cohort at 2-year follow-up was 27 total patients compared with 32 patients with BPTB. However, our initial pre-study power analysis determined a minimum of 24 patients was necessary in each group to detect a clinically relevant difference in KT-1000 knee AP laxity side-to-side difference (compared with normal, contralateral knee) of 2.5 mm between the 2 groups, assuming a standard deviation of 3.0.

One final limitation is the fact that the BPTB group ($n = 9$) had more partial meniscectomies done than the AIST group ($n = 1$). However, data analysis showed no

differences in either patient objective or subjective outcome measures when comparing individuals who received treatment for meniscal pathology and individuals who presented with uninjured menisci. Furthermore, all partial meniscectomies were performed on the lateral meniscus, which has been found to have no impact on knee laxity clinically in the short term.¹³²⁻¹³⁴

Conclusions

ACL reconstruction with an all-inside quadrupled semitendinosus autograft construct was equivalent to patellar BPTB autograft based on KT-1000 stability testing in athletes 24 years or younger.

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